

NMOS CHARACTERIZATION

Cadence Virtuoso / Spectre simulation

STUDENT

MUHAMMED HADI M

ROLL NO.

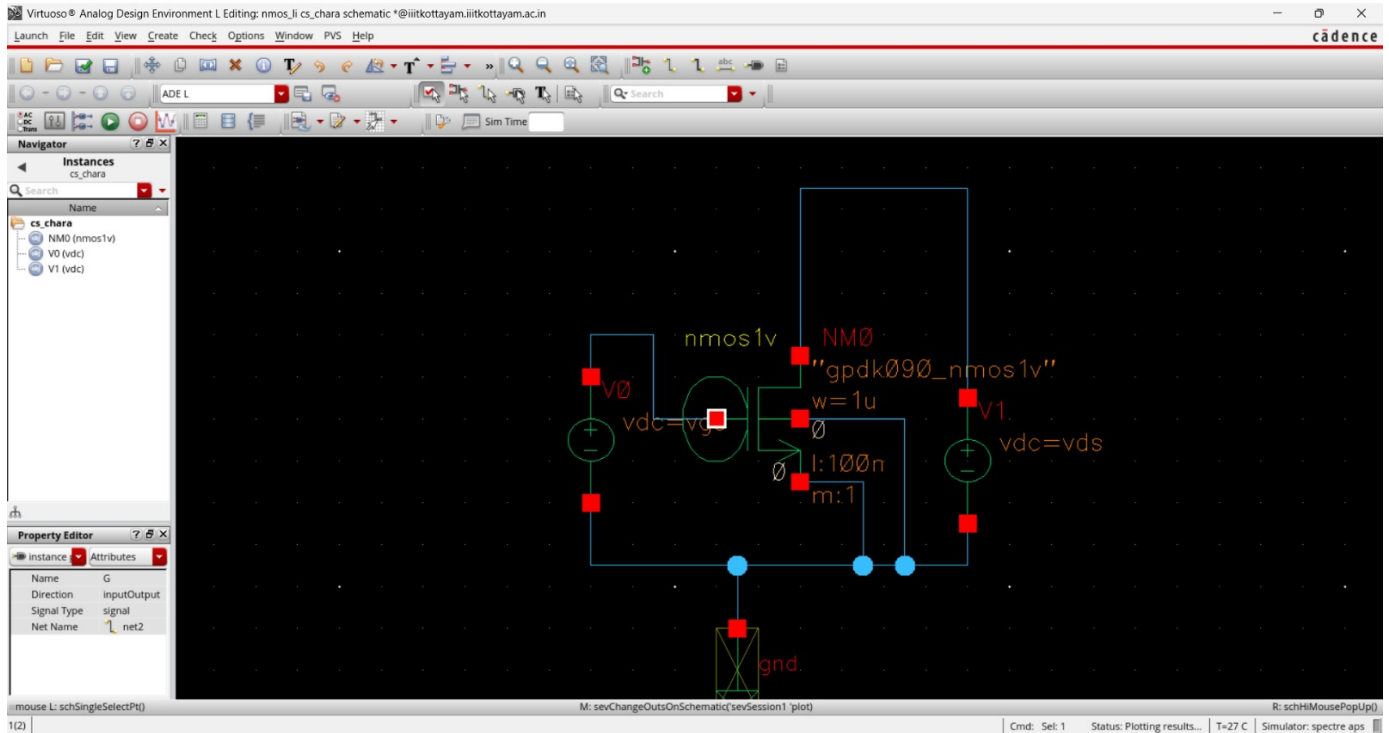
2025BEC0060

REPOSITORY

https://github.com/mhdhadi055-png/AnalogElectronicsLab_2025BEC0060

Schematic

NMOS test circuit



NMOS characterization test circuit implemented in Cadence Virtuoso.

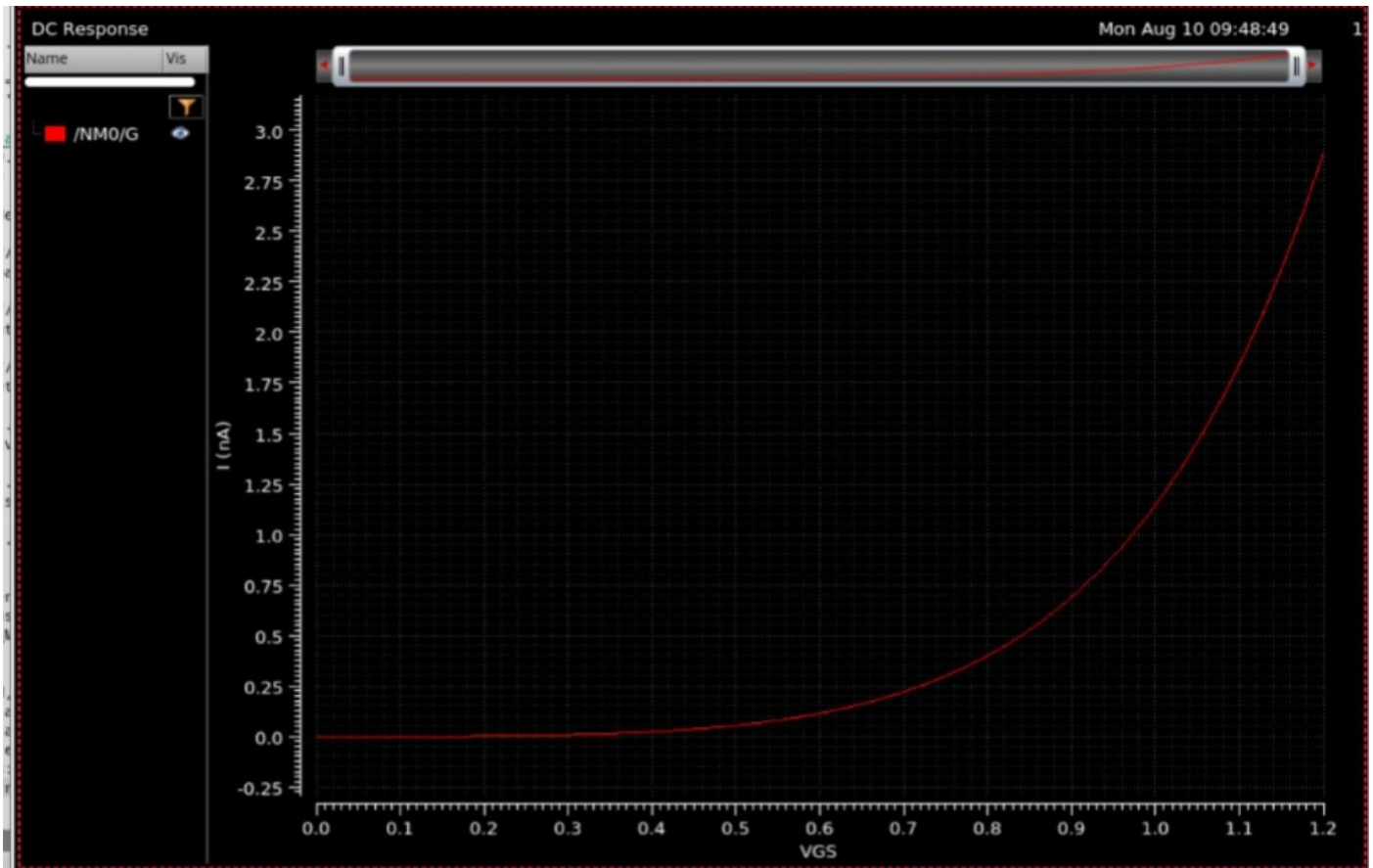
Device dimensions: $W = 1 \mu\text{m}$, $L = 100 \text{ nm}$. VGS and VDS are applied through the voltage sources.

PARAMETRIC SWEEP

VGS sweep range: 0 V to 1.2 V

VGS Transfer Characteristic

Drain current response versus gate-source voltage



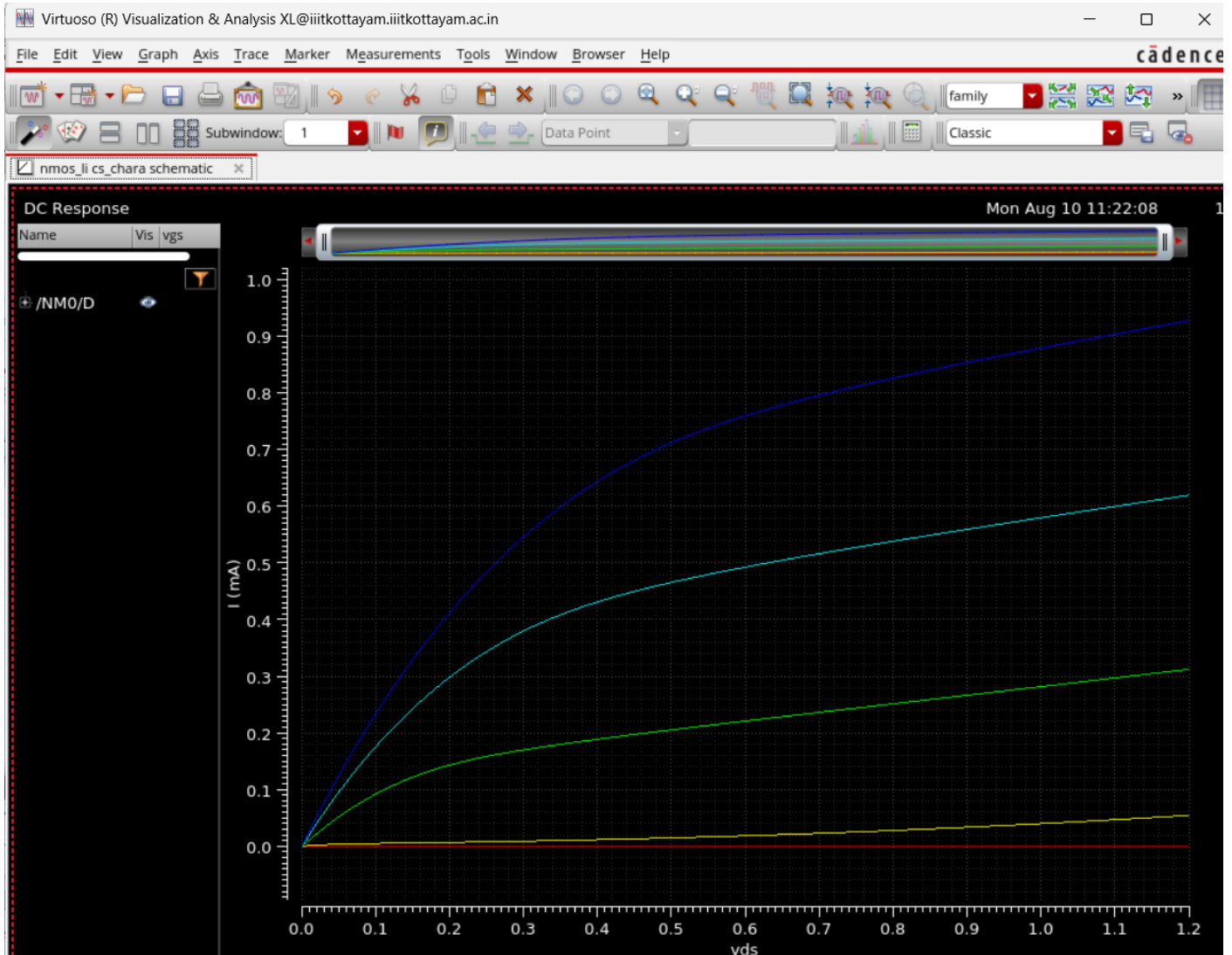
DC response showing drain current increasing as V_{GS} is increased.

OBSERVATION

The current remains very small at low gate-source voltage and rises strongly as V_{GS} increases.

VDS Output Characteristics

Parametric sweep for multiple VGS values



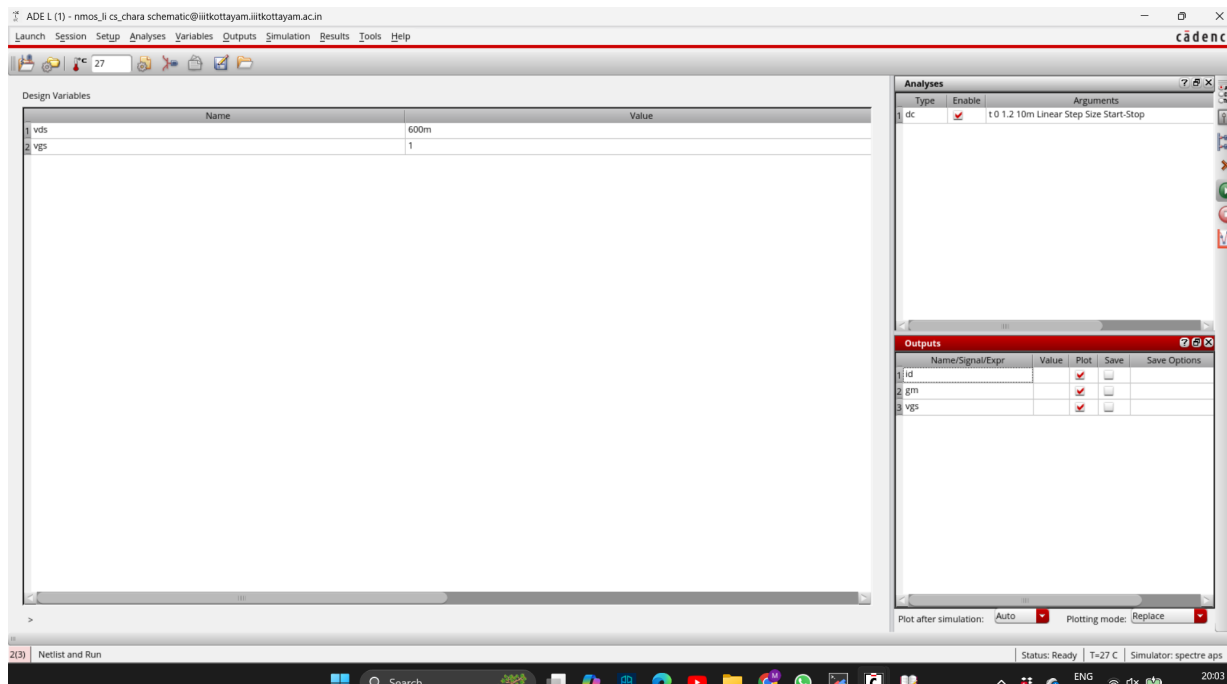
Drain current I_D plotted against V_{DS} for multiple V_{GS} values.

OBSERVATION

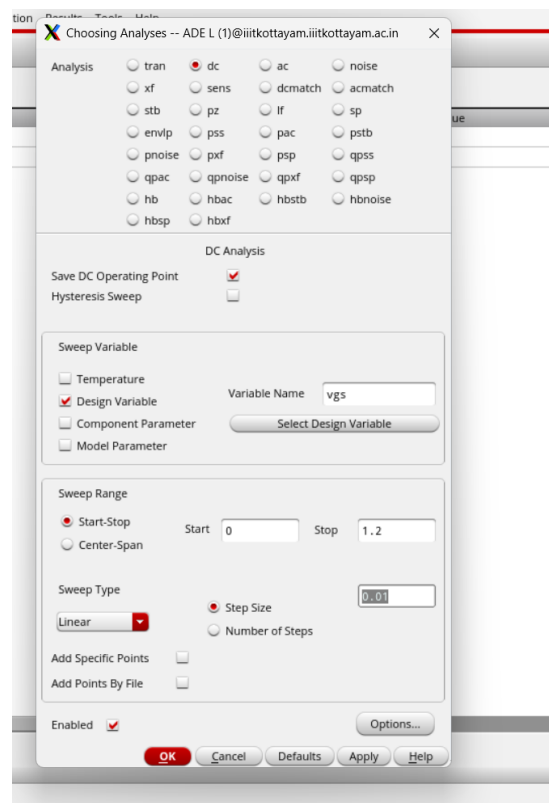
Higher V_{GS} produces higher drain current, while the curves become less steep as V_{DS} increases.

Simulation Setup

DC analysis configuration and design variables



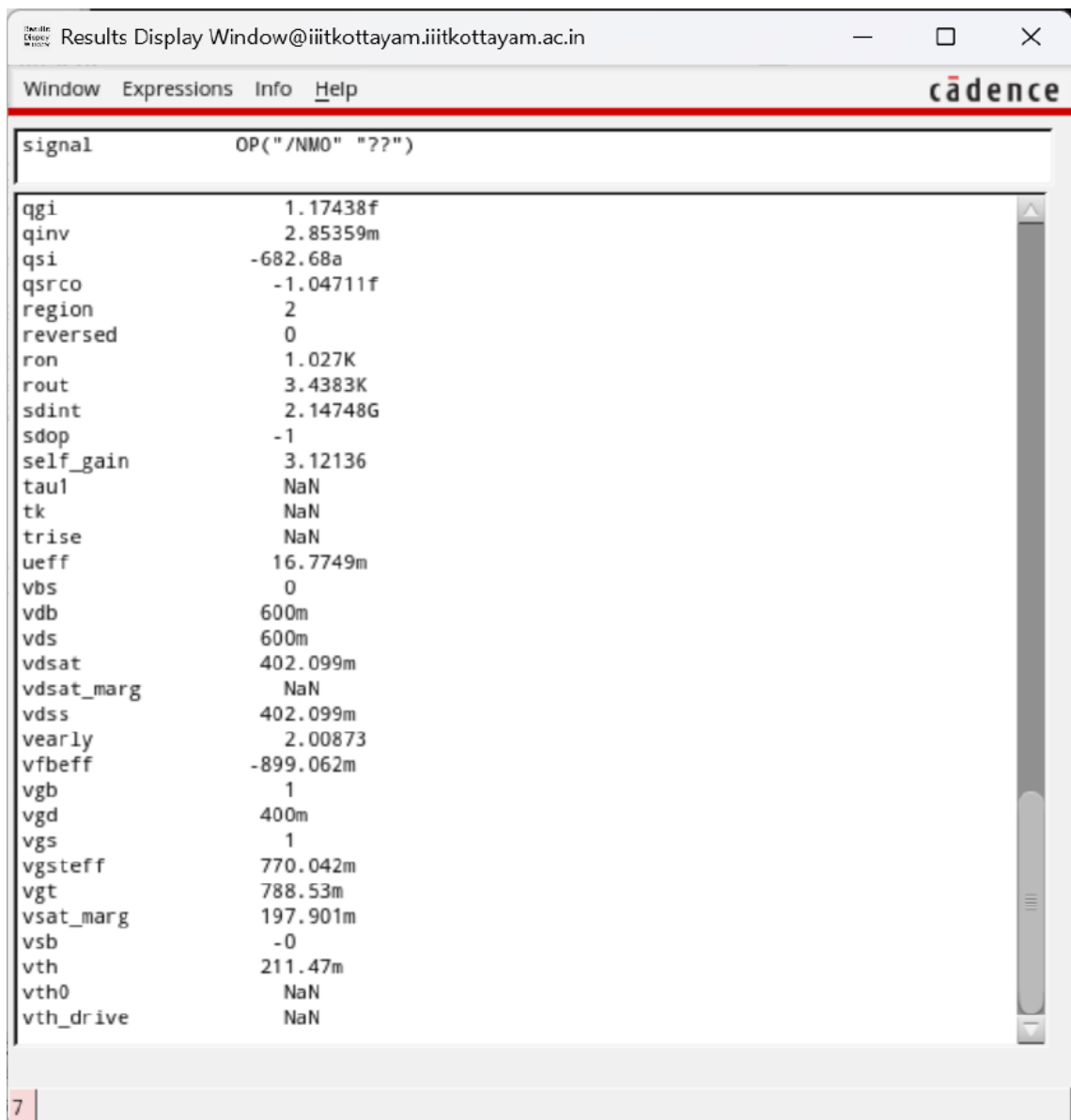
ADE L setup showing VDS and VGS design variables and the enabled DC analysis.



DC analysis configuration with VGS swept from 0 V to 1.2 V.

DC Operating Point

Spectre operating-point results



signal	OP("/NMO" "?")
qgi	1.17438f
qinv	2.85359m
qsi	-682.68a
qsrco	-1.04711f
region	2
reversed	0
ron	1.027K
rout	3.4383K
sdint	2.14748G
sdop	-1
self_gain	3.12136
taul	NaN
tk	NaN
trise	NaN
ueff	16.7749m
vbs	0
vdb	600m
vds	600m
vdsat	402.099m
vdsat_marg	NaN
vdss	402.099m
vearly	2.00873
vfbeff	-899.062m
vgb	1
vgd	400m
vgs	1
vgsteff	770.042m
vgt	788.53m
vsat_marg	197.901m
vsb	-0
vth	211.47m
vth0	NaN
vth_drive	NaN

Spectre operating-point result window for the NMOS device.

KEY RESULT

VDS = 0.6 V • VGS = 1 V

Reported threshold voltage: approximately 211.47 mV.